

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CE/CS/EC/EE/IT/ME/SE

19-200-0401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

(2019 Scheme)

Maximum Marks: 60

Time: 3 Hours

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Transform a region to another using conformal mapping.
 CO2: Evaluate real integrals using Residue theorem.
 CO3: Form and solve partial differential Equations.
 CO4: Determine solution of partial differential equation for vibrating string and heat conduction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic.	3	L3	1	2	
(b) Find the image of $ z - 1 = 1$ under the mapping $w = 1/z$.	3	L2	1	2	
(c) Using Cauchy's integral formula, evaluate $\int_c \frac{e^{-z}}{z^2} dz$.	3	L3	2	3	
(d) Expand $\frac{1}{z^2 - 3z + 2}$ in the region $1 < z < 2$.	3	L3	2	3	
(e) Solve $p^2 = p + q$.	3	L2	3	2	
(f) Solve $z = p^2 + q^2$.	3	L2	3	2	
(g) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$; given $u(x, 0) = 6e^{-3x}$.	3	L2	4	2	
(h) Find the solution of the one dimensional wave equation.	3	L1	4	1	

PART B

(4 × 12 = 48)

II. (a)	Show that $f(z) = \sqrt{ XY }$ is not analytic at the origin, although C-R equations are satisfied at that point.	6	L3	1	3
(b)	Find the analytic function $f(z)$ whose real part is $e^{2x}(x \cos 2y - y \sin 2y)$.	6	L3	1	3
OR					
III. (a)	Determine the analytic function $w = u + iv$ where $v = \log(x^2 + y^2) + x - 2y$.	6	L3	1	3
(b)	Find the bilinear transformation which maps the points $z = 2, i, -2$ in to points $w = 1, i, -1$.	6	L2	1	2

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Verify Cauchy's theorem for the integral of z^3 taken over the boundary of the rectangle with vertices $-1, 1, 1+i, -1+i$.	6	L2	2	2
	(b) Expand $\frac{z-1}{z^2}$ in powers of $z-1$ and determine the region of convergence.	6	L3	2	3
OR					
V.	(a) Using Cauchy's Residue Theorem, evaluate $\int_c \frac{z}{(z-1)(z-2)^2} dz$ where c is $ z-2 = \frac{1}{2}$.	6	L3	2	3
	(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$.	6	L2	2	2
VI.	(a) Find the partial differential equation of all spheres whose centres lie on z -axis.	6	L3	3	2
	(b) Solve $r - 4s + 4t = \sin(3x + 4y)$.	6	L3	3	3
OR					
VII.	(a) Find the partial differential equation by eliminating the arbitrary function from $f(x+y+z, x^2+y^2+z^2) = 0$.	6	L2	3	2
	(b) Solve $x(y-z)p + y(z-x)q = z(x-y)$.	6	L2	3	3
VIII.	(a) Solve $U_{xx} - U_y = 0$ by the method of separation of variables.	6	L1	4	1
	(b) A string is stretched and fastened to two points 1 apart. Motion is started by displacing the string in the form $y = a \sin\left(\frac{\pi x}{l}\right)$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{\pi ct}{l}\right)$.	6	L4	4	4
OR					
IX.	(a) Solve the heat equation.	6	L1	4	1
	(b) Solve Laplace's equation in two dimensions.	6	L3	4	3

Blooms's Taxonomy Level's

L1 - 15%, L2 - 45%, L3 - 25%, L5 - 15%.
